

Boise State University - Project Management Plan

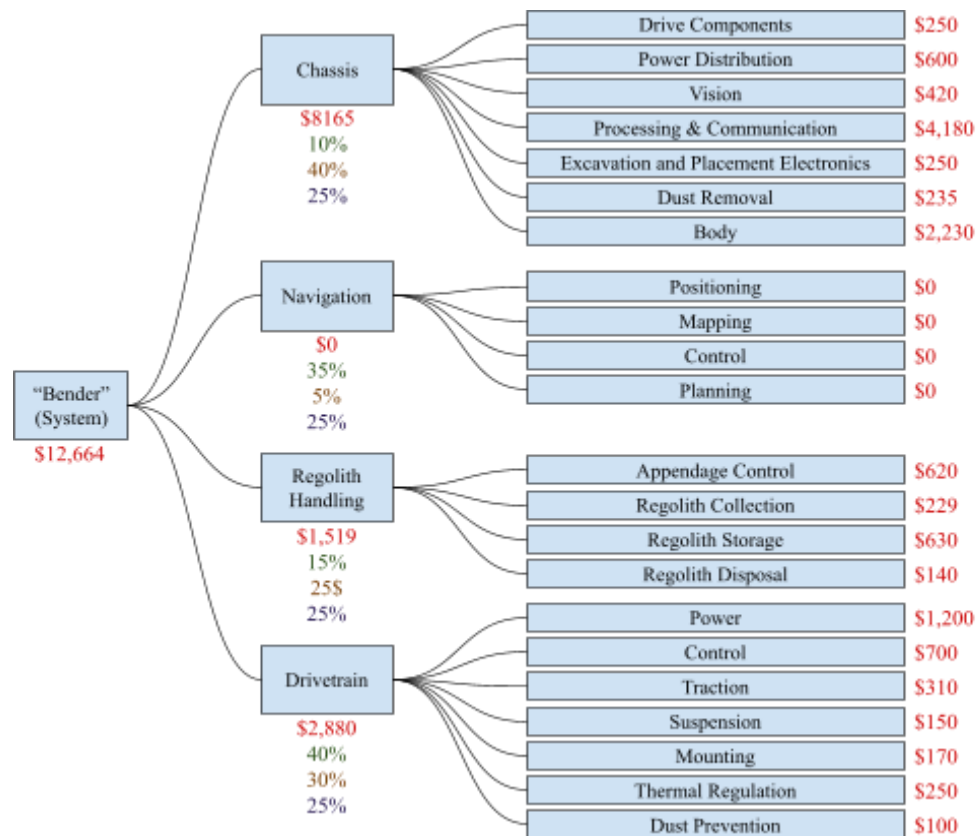
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Executive Summary

The Project Management Plan for Boise State University outlines a strategic framework for developing the lunar vehicle, “Bender”. This project aims to design and construct a vehicle capable of navigating simulated lunar terrain, handling regolith, and performing autonomous tasks. The project is structured into four primary subsystems: chassis, drivetrain, navigation, and regolith handling, each with associated costs, power/time distribution, and weight metrics. This project's key technical objectives are maximizing automation through path navigation and object identification, including collecting and depositing regolith while maintaining system functionality. Risk mitigation strategies were developed with contingency plans for motor, software, and other technical failures. This comprehensive management plan aims to deliver a highly functional autonomous lunar vehicle while adhering to established performance targets and budgets, paving the way for advancements in lunar exploration.

System Hierarchy



Above is a diagram of Bender’s system hierarchy broken down into its four subsystems and their corresponding nested components. Anticipated costs are highlighted in red for each subsystem. The estimated power distribution is highlighted in green. Weight distribution for each system is highlighted in orange. The percentage of work across teams is highlighted in blue, representing an even time distribution of tasks across all subsystems. The team plans to review our TPM budgets monthly and constantly update the subsystem allocations accordingly.

Project Technical Objectives

The team's primary Technical Objectives are as follows: first, maximize automation of the vehicle. This necessitates that the vehicle be able to plan and traverse a path across simulated lunar terrain to a desired location while avoiding obstacles without human intervention. Secondly, the vehicle will maintain mobility and move under its power within the simulated lunar environment. Thirdly, the vehicle will collect regolith, transport it, and deposit it into a berm while minimizing the unintentional loss of the regolith and dust contamination to preserve functionality.

Initial Technical Performance Measures

<i>Technical Performance Measures</i>	<i>Target</i>	<i>Related System Hierarchy</i>	<i>Critical Schedule Milestones and Date</i>
Nav Power	Navigational computation shall take no more than 60 Watts	1.1.0 Navigation	Critical Design Review 01/14/2025
Chassis Weight	The chassis of the vehicle shall have a mass of 30 kg or less	1.2.0 Chassis	Critical Design Review 01/14/2025
Regolith System Weight	The regolith handling subsystem of the vehicle shall have an unladen mass of 10 kg	1.3.0 Regolith Handling	Critical Design Review 01/14/2025
Positional Accuracy	The navigation system shall be able to locate the vehicle's position relative to a global reference point with an accuracy of ± 5 cm	1.1.1 Positioning	Systems Engineering Paper 04/09/2025
Compute Time	The vehicle shall take no more than 500 ms to compute terrain data, perform mapping, and determine position relative to the global reference point.	1.1.0 Navigation	Systems Engineering Paper 04/09/2025
Terrain Traversal	The vehicle shall be able to traverse the simulated lunar environment while transporting 10 kg of regolith without losing locomotive capability	1.4.3 Traction	Systems Engineering Paper 04/09/2025
Drivetrain Weight	The drivetrain assembly shall have a mass of 15 kg or less	1.4.0 Drivetrain	Critical Design Review 01/14/2025
Drivetrain Power	The drivetrain shall use no more than 400 watts at full capacity	1.4.1 Power	Critical Design Review 01/14/2025
Turning Performance	The vehicle shall be able to make a 180-degree change in direction within a radius of 2 meters	1.4.2 Control	Critical Design Review 01/14/2025
Regolith Management	The vehicle shall excavate, move, and deposit regolith within the target berm area of 1.5m x 0.9 m (UCF) and 2.2 m x 0.9 m (KSC)	1.3.0 Regolith Handling	Critical Design Review 01/14/2025
Regolith Power	The regolith subsystem shall not exceed 200 Watts	1.3.1 Appendage Control	Critical Design Review 01/14/2025
Dimension	The vehicle consisting of chassis, locomotion method, electronics, and regolith transport mechanism shall fit within a volume of 1.5 x 0.75 x 0.75 (m)	1.0.0 Bender	Preliminary Design Review 11/12/2024

Lifting Points	The vehicle shall have 4 lifting points distanced from the center of mass so that each splits the load equally. No single handle shall carry more than 20kg when lifted. All handles shall be clearly marked with (ISO 7000-1368)	1.2.7 Body	100% Completion of Build 03/22/2025
E-stop	The vehicle shall have an unmodified "Commercial Off-The-Shelf" (COTS) red button, that is readily available to the operator following OSHA's IEC 60204-1 standard	1.2.2 Power Distribution	100% Completion of Build 03/22/2025
Mapping	The vehicle shall be able to map the surrounding terrain and detect obstacles with accuracy within ± 5 cm.	1.1.2 Mapping	Systems Engineering Paper 04/09/2025

Project Resources & Budget

Overall Project Cost: \$23156					
Navigation	\$0.00		Chassis	\$8,165.00	
	Positioning	\$0.00		Drive Components	\$250.00
	Mapping	\$0.00		Power distribution	\$600.00
	Control	\$0.00		Vision	\$420.00
	Planning	\$0.00		Processing & Communication	\$4,180.00
Regolith Handling	\$1,619.00			Excavation and Placement Electronics	\$250.00
	Appendage Control	\$620.00		Dust Removal	\$235.00
	Regolith Collection	\$229.00		Body	\$2,230.00
	Regolith Storage	\$630.00	Travel	\$13,372.00	
	Regolith Disposal	\$140.00		Flights	\$3,000.00
Drivetrain	\$2,880.00			Hotel	\$4,222.00
	Power	\$1,200.00		Transportation	\$1,450.00
	Control	\$700.00		Shipping	\$800.00
	Traction	\$310.00		Registration	\$0.00
	Suspension	\$150.00		Stipend	\$3,900.00
	Mounting	\$170.00		# Of Travelers	6
	Thermal Regulation	\$250.00			
	Dust Prevention	\$100.00			

The team plans to review our budget monthly and constantly update our income and expenses to adjust our budget and spending accordingly. This will allow the team to have a tight grasp on our financials and potentially allow more travelers to the competition. All hardware for

10/03/2024

navigation has been included in the Chassis budget and there are no software expenses for the Navigation subsystem. Additionally, there are three main budget milestones where the team plans to make their large purchases. The “First Wave of Part Orders” will take place on 11/19/2024 after our PRD and PMP Review with our Industry Advisory Board. The “Second Wave of Part Orders” will take place on 12/10/2024 after our initial prototypes have been built and tested. Lastly, the “Third Wave of Part Orders” which will be our last major order will take place on 2/3/2025 after our CDR as the team works towards our 100% “Completion of Build Milestone”

Project Schedule

To keep the project on track and adaptable, the team will implement a change management process and conduct monthly scheduled reviews during the last Tuesday of each month. These reviews will address potential issues early, such as scope changes or resource constraints, and identify opportunities to accelerate tasks if possible. The team will prioritize any critical tasks at risk of delay, redistributing time to ensure the project stays on course.

Milestones, Projects, and Phases	8/25/2024	9/1/2024	9/8/2024	9/15/2024	9/22/2024	9/29/2024	10/6/2024	10/13/2024	10/20/2024	10/27/2024	11/3/2024	11/10/2024	11/17/2024	11/24/2024	12/1/2024	12/8/2024	12/15/2024	12/22/2024	12/29/2024	1/5/2025	1/12/2025	1/19/2025	1/26/2025	2/2/2025	2/9/2025	2/16/2025	2/23/2025	3/2/2025	3/9/2025	3/16/2025	3/23/2025	3/30/2025	4/6/2025	4/13/2025	4/20/2025	4/27/2025	5/4/2025	5/11/2025	5/18/2025	5/25/2025	
Ph) Initiating																																									
M) Team Application																																									
M) Team Selections Released																																									
M) Registration for Selected Teams																																									
Ph) Planning																																									
M) System Requirements Review																																									
M) Preliminary Design Review																																									
M) First Wave of Part Orders																																									
Ph) Executing																																									
M) Intial Prototypes																																									
M) Second Wave of Part Orders																																									
M) 60% Completion of build																																									
M) Critical Design Review																																									
M) System integration Testing																																									
M) 100% Completion of Build																																									
Ph) Monitoring/ Controlling																																									
M) STEM Engagement Report & Team Photos																																									
M) Presentation and Demonstration																																									
M) Systems Engineering Paper																																									
M) Proof of Life Video Link																																									
M) Round 2 Pass/ Fail Anoucements																																									
M) Student & Faculty Registration for On-Site Comp																																									
M) UCF’s Lunabotics Qualification																																									
M) NASA’s Lunabotics Challenge																																									
Ph) Closing																																									
M) Completion & Disposal of Project																																									